

PHASE 7 – PROJECT DOCUMENTATION

1. INTRODUCTION

The Project Documentation phase presents the complete documentation of the **Import Data using Transform Maps (Spreadsheet)** project developed in the ServiceNow environment. This phase consolidates the project requirements, objectives, system design, development activities, data transformation process, testing results, and final project outcomes into a structured document.

The project demonstrates how structured employee data maintained in an external spreadsheet can be imported into ServiceNow using **Import Sets and Transform Maps**. The spreadsheet data is first loaded into an Import Set table, which acts as a temporary staging area. A Transform Map is then configured to transfer the source data into the appropriate fields of the Employee Test target table.

The project also demonstrates field mapping, data transformation, record validation, and Coalesce configuration. Coalesce helps identify existing records based on the Employee ID and prevents unnecessary duplicate records when the same employee data is imported again. The complete process provides a practical understanding of how bulk data can be efficiently transferred and managed within ServiceNow.

2. PROJECT TITLE

Import Data using Transform Maps (Spreadsheet)

3. PROJECT OVERVIEW

The project implements a structured spreadsheet data import process in ServiceNow using **Import Sets and Transform Maps**.

Employee information is initially prepared in a spreadsheet containing details such as:

- Employee ID
- Employee Name
- Email
- Department

- Location

The spreadsheet is downloaded in Excel format and uploaded into ServiceNow through the **Load Data** functionality. The uploaded information is stored in the Employee Import table, which acts as the staging table for the imported data.

A custom **Employee Test** table is created as the target table. A Transform Map named **Sample Spreadsheet Import** is then configured between the Employee Import source table and the Employee Test target table.

The source fields are mapped with the corresponding target fields, and the transformation is executed. After transformation, the employee records are available in the Employee Test table.

The project also uses **Coalesce** with Employee ID to identify existing records. This allows existing employee information to be updated when new values are imported instead of creating duplicate records.

4. PROBLEM STATEMENT

In many organizations, employee information may be maintained in external spreadsheets. Manually entering a large number of employee records into ServiceNow can be time-consuming and may lead to data entry errors.

When employee information is received in spreadsheet format, there is a need for an efficient method to transfer the data into ServiceNow. The imported data must also be stored in the correct fields and verified after the import process.

Another important issue is duplicate data. If the same employee information is imported multiple times, the system should be able to identify existing records instead of creating unnecessary duplicates.

Therefore, a structured data import mechanism is required to:

- Load spreadsheet data into ServiceNow.
- Store imported data temporarily in an Import Set table.
- Map source fields to the correct target fields.
- Transform the imported data into the target table.

- Verify the imported records.
- Update existing records when required.
- Prevent unnecessary duplicate records.

This project addresses these requirements using ServiceNow Import Sets, Transform Maps, field mapping, and Coalesce.

5. PROJECT OBJECTIVES

The main objectives of the project are:

- To understand the spreadsheet data import process in ServiceNow.
- To create a custom Employee Test table.
- To create the required employee fields.
- To prepare structured employee data in spreadsheet format.
- To create an Import Set table for loading external data.
- To create a Transform Map between the source and target tables.
- To map source fields with the corresponding target fields.
- To execute the transformation process successfully.
- To verify the imported employee records.
- To configure Coalesce for preventing duplicate records.
- To update existing employee records when their details are changed.

6. REQUIREMENT ANALYSIS

6.1 Data Requirements

The project requires employee information in spreadsheet format.

The spreadsheet contains the following fields:

Field	Description
Employee ID	Unique identification value of the employee

Field	Description
Name	Name of the employee
Email	Email address of the employee
Department	Department associated with the employee
Location	Location of the employee

The sample employee data used in the project is:

Employee ID	Name	Email	Department	Location
SB0001	User1	user1@gmail.com	Servicenow	Tamil Nadu
SB0002	User2	user2@gmail.com	Servicenow	Tamil Nadu
SB0003	User3	user3@gmail.com	Servicenow	Tamil Nadu
SB0004	User4	user4@gmail.com	Servicenow	Tamil Nadu
SB0005	User5	user5@gmail.com	Servicenow	Tamil Nadu

6.2 Target Table Requirements

The target table created in ServiceNow is:

Employee Test

Table Name u_employee_test

The table contains the required fields for storing the employee information.

6.3 Field Requirements

Field	Type
Employee ID	String
Employee Name	String
Email	String

Field	Type
Department	String
Location	String

6.4 Import Set Requirements

The Import Set table created temporarily store the spreadsheet data before transformation.

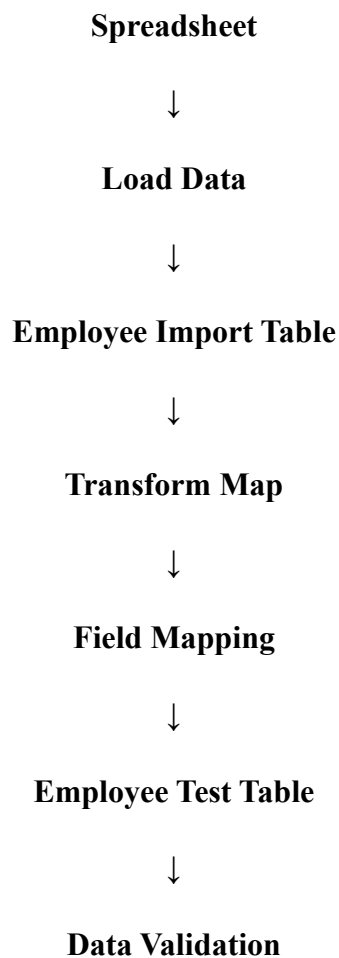
Label: Employee Import

Name: u_employee_import

7. SYSTEM DESIGN

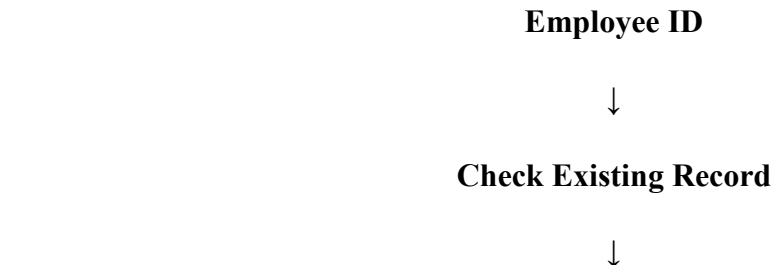
The system follows a simple source-to-target data transformation model.

7.1 Data Import Flow



The spreadsheet acts as the external source of employee information. The Employee Import table temporarily stores the uploaded data. The Transform Map processes the source data and transfers it into the Employee Test target table.

7.2 Coalesce Flow



Existing Employee ID?

→ **Yes:** Update existing record

→ **No:** Insert new record

This process helps prevent duplicate employee records during repeated imports.

8. PROJECT IMPLEMENTATION

8.1 Employee Test Table Creation

A custom Employee Test table was created in ServiceNow using the Tables module. The table acts as the target table for the imported employee information.

The table was created with the following details:

- Label: Employee Test
- Name: u_employee_test

8.2 Employee Field Creation

The required fields were added to the Employee Test table.

The fields include:

- Employee ID
- Employee Name
- Email

- Department
- Location

These fields correspond to the information available in the employee spreadsheet.

8.3 Spreadsheet Preparation

A Google Spreadsheet was created with five employee records.

The employee data contains Employee IDs from **SB0001 to SB0005**, along with their names, email addresses, department, and location.

The spreadsheet was downloaded in Excel format for use in ServiceNow.

8.4 Import Set Creation

The spreadsheet data was loaded using the ServiceNow **Load Data** functionality. The Employee Import table was created as the Import Set table. The table temporarily stores the spreadsheet information before the transformation process.

8.5 Transform Map Creation

A Transform Map named **Sample Spreadsheet Import** was created.

The source table was selected as **Employee Import**, while the target table was selected as **Employee Test**.

The Transform Map provides the connection between the imported spreadsheet data and the target employee records.

8.6 Field Mapping

The source and target fields were mapped according to their corresponding values.

Source Field	Target Field
Employee ID	Employee ID
Name	Employee Name
Email	Email

Source Field	Target Field
Department	Department
Location	Location

The mapping ensures that the imported values are stored in the correct fields of the Employee Test table.

8.7 Data Transformation

After completing the field mapping, the Transform Map was executed.

The source records from the Employee Import table were processed and transferred to the Employee Test table.

The transformation process successfully created the employee records in the target table.

8.8 Coalesce Configuration

The Employee ID field was configured as the Coalesce field.

Coalesce helps ServiceNow determine whether the imported Employee ID already exists in the target table.

If the Employee ID does not exist, a new record is inserted.

If the Employee ID already exists, the existing record is identified and updated with the latest values.

This process helps maintain data consistency and reduces duplicate records.

9. PROJECT TESTING

The implemented data import process was tested to verify that the system performs according to the project requirements.

The testing included:

- Loading spreadsheet data into the Import Set table.
- Executing the Transform Map.
- Checking the target Employee Test table.

- Verifying source-to-target field mapping.
- Testing new employee record insertion.
- Testing existing employee record updates.
- Testing duplicate data handling using Coalesce.
- Checking Transform History for the final transformation status.

The testing confirmed that the employee information was successfully transferred from the spreadsheet into the ServiceNow target table.

10. TEST RESULTS

The following test cases were used to verify the implementation:

Test Case	Action	Actual Result	Status
TC01	Upload spreadsheet	Employee data was loaded successfully	Pass
TC02	Execute Transform Map	Employee records were created successfully	Pass
TC03	Verify field mapping	Values appeared in correct target fields	Pass
TC04	Import new Employee ID	New record was inserted	Pass
TC05	Import existing Employee ID with changed details	Existing record was updated	Pass
TC06	Import unchanged data again	Duplicate records were not created	Pass

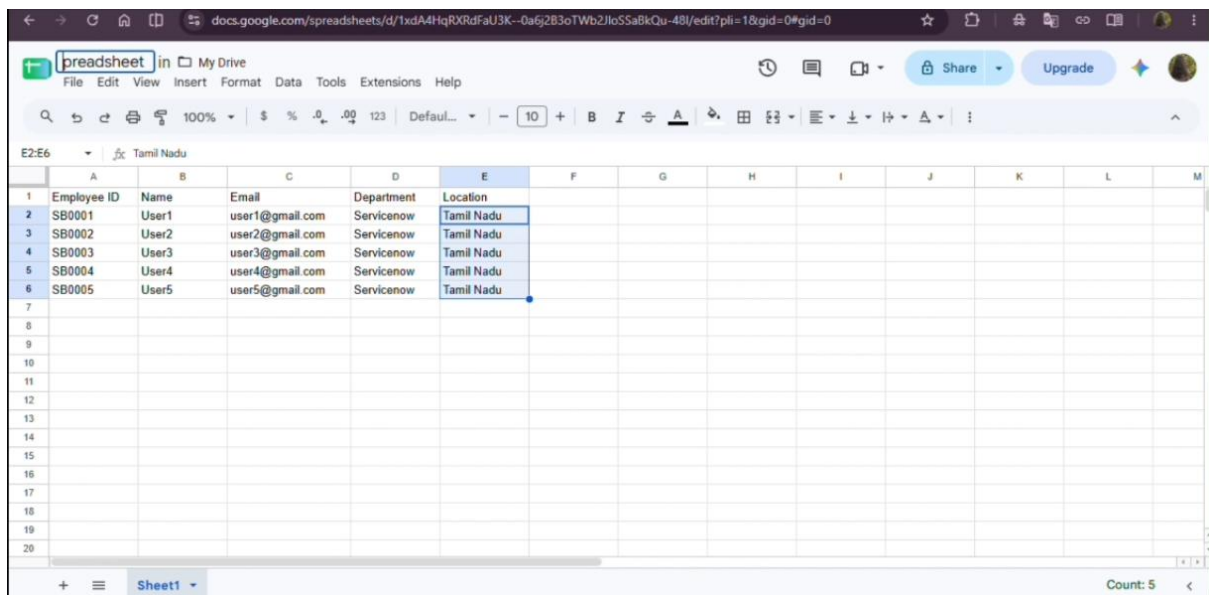
All the documented test cases produced the expected results.

11. SCREENSHOT AND EVIDENCE DOCUMENTATION

The project development activities were supported with screenshots captured during implementation.

Development Evidence

The following screenshots were captured

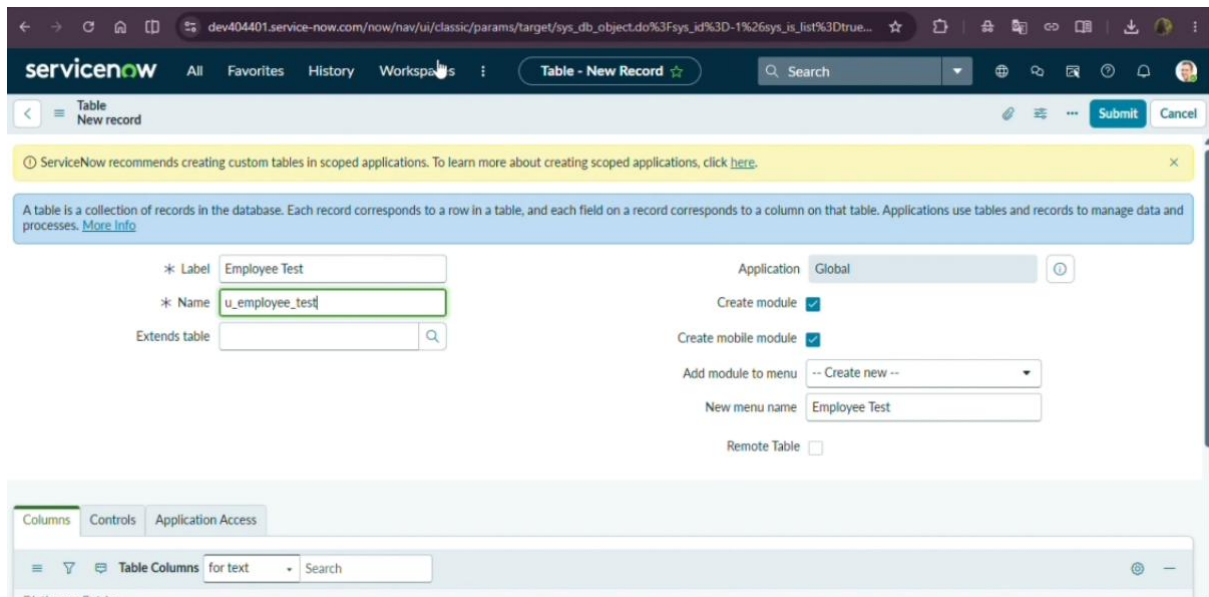


A screenshot of a Google Spreadsheet titled 'breadsheet' in 'My Drive'. The spreadsheet has columns A through M and rows 1 through 20. The data is as follows:

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Employee ID	Name	Email	Department	Location								
2	SB0001	User1	user1@gmail.com	Servicenow	Tamil Nadu								
3	SB0002	User2	user2@gmail.com	Servicenow	Tamil Nadu								
4	SB0003	User3	user3@gmail.com	Servicenow	Tamil Nadu								
5	SB0004	User4	user4@gmail.com	Servicenow	Tamil Nadu								
6	SB0005	User5	user5@gmail.com	Servicenow	Tamil Nadu								
7													
8													
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12													
13													
14													
15													
16													
17													
18													
19													
20													

The bottom of the spreadsheet shows a 'Sheet1' tab and a 'Count: 5' indicator.

Figure 7.2 – Employee Test table fields



A screenshot of the ServiceNow 'Table - New Record' form. The form is titled 'Table - New Record' and has a 'Submit' button. It contains the following fields:

- * Label: Employee Test
- * Name: u_employee_test
- Extends table: (empty)
- Application: Global
- Create module: ☒
- Create mobile module: ☒
- Add module to menu: -- Create new --
- New menu name: Employee Test
- Remote Table: ☐

Below the form, there are tabs for 'Columns', 'Controls', and 'Application Access'. The 'Columns' tab is selected, showing a 'Table Columns' section with a search bar.

Figure 7.2 – Employee Test table fields

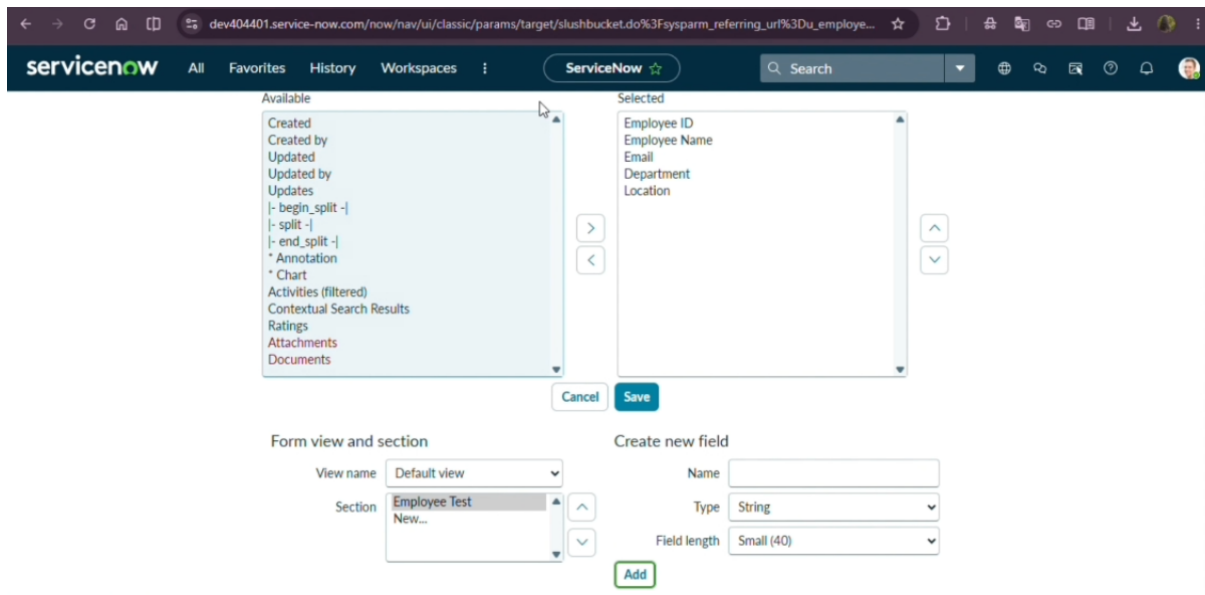


Figure 7.3 – Sample employee data in Google Spreadsheet

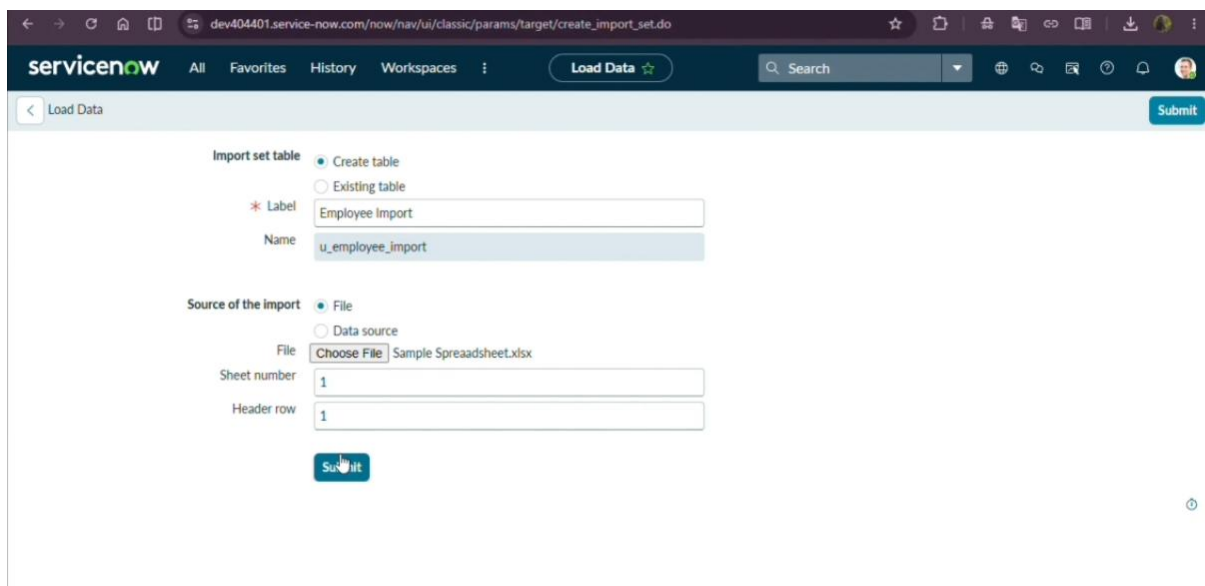


Figure 7.4 – Employee Import table creation

dev404401.service-now.com/now/nav/ui/classic/params/target/sys_transform_map.do%3Fsys_id%3D-1%26sysparm_que...

servicenow All Favorites History Table Transform Map - New Record Search

Table Transform Map
New record

* Name: Sample Spreadsheet Import

* Source table: Employee Import [u_employee_im...]

Active: ☒

Run business rules: ☒

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty coalesce fields: ☐

Application: Global

Created:

* Target table: Employee Test [u_employee_test]

Order: 100

Run script: ☐

Submit

Related Links
[Auto Map Matching Fields](#)
[Mapping Assist](#)

Figure 7.5 – Transform Map creation

dev404401.service-now.com/now/nav/ui/classic/params/target/mapping_assist.do%3Fsysparm_stack%3Dno%26sysparm...

servicenow All Favorites History Workspaces ServiceNow Search

Mapping Assist Save Cancel

Source: Employee Import

Comment
Created
Created by
Department
Email
Employee ID
Error
Import set run
Location
Name

Add: > Remove: <

Field Map	
Name	Employee Name
Email	Email
Employee ID	Employee ID
Location	Location
Department	Department

Remove: > Add: <

Target: Employee Test

Save Cancel

Data Viewer

Show: ☒ All Fields ☐ Mapped Fields

Employee Import Viewing: 1

Field	Value
-------	-------

Employee Test Viewing: 1

Field	Value
-------	-------

Figure 7.6 – Source and target field mapping

Employee ID	Employee Name	Department	Email	Location
SB0004	User4	ServiceNow	user4@gmail.com	Tamil Nadu
SB0001	User1	ServiceNow	user1@gmail.com	Tamil Nadu
SB0003	User3	ServiceNow	user3@gmail.com	Tamil Nadu
SB0002	User2	ServiceNow	user2@gmail.com	Tamil Nadu
SB0006	User6	ServiceNow	user6@gmail.com	Tamil Nadu
SB0005	User5	ServiceNow	user5@gmail.com	Tamil Nadu

Figure 7.7 – Transform History and final employee records

These screenshots provide evidence of the main development activities performed during the implementation of the project. No separate screenshots were captured during the Project Testing phase. The testing activities and results are documented using the test cases and result table.

12. PROJECT OUTCOME

The project successfully demonstrates the process of importing structured spreadsheet data into ServiceNow using Import Sets and Transform Maps.

The final system provides:

- Successful spreadsheet data loading.
- Employee Import staging table.
- Employee Test target table.
- Correct source-to-target field mapping.
- Successful data transformation.
- Verification of imported employee records.
- Coalesce-based duplicate prevention.

- Updating of existing employee records.
- Insertion of new employee records.
- Identification of unchanged records during repeated imports.
- Transform History for verifying transformation results.

The project provides a practical understanding of how external spreadsheet data can be migrated into ServiceNow in an organized and controlled manner.

13. CHALLENGES ENCOUNTERED

During the implementation of the project, several areas required careful configuration and verification.

The main challenges included:

- Creating the Employee Test table with the required fields.
- Ensuring that the spreadsheet columns matched the required target fields.
- Creating the Employee Import table correctly.
- Selecting the correct source and target tables in the Transform Map.
- Mapping the source fields with the appropriate target fields.
- Verifying the transformation results.
- Understanding the difference between inserted, updated, and ignored records.
- Configuring Coalesce correctly to identify existing Employee IDs.
- Checking repeated imports to ensure duplicate records were not created.
- Verifying the final records in the Employee Test table.

These challenges were addressed by checking each stage of the import and transformation process and verifying the results after every major operation.

14. FUTURE ENHANCEMENTS

The project can be enhanced in the future by:

- Adding more employee fields such as phone number, designation, and joining date.

- Supporting additional spreadsheet formats.
- Adding data validation before importing records.
- Adding validation for duplicate Employee IDs in the source spreadsheet.
- Implementing more advanced field transformations.
- Adding data cleansing rules for incomplete or incorrect records.
- Extending the Transform Map to support larger employee datasets.
- Adding automated scheduled data imports.
- Creating reports based on imported employee information.
- Adding additional validation and error-handling mechanisms.
- Extending the process to other ServiceNow tables and business data.
- These enhancements can make the data import process more flexible and suitable for larger real-world data migration requirements.

15. CONCLUSION

The **Import Data using Transform Maps (Spreadsheet)** project demonstrates a complete data import and transformation process in ServiceNow. The project begins with preparing structured employee information in a spreadsheet and continues with loading the data into an Import Set table.

A custom Employee Test table was created as the target table, and the required employee fields were configured. A Transform Map was then created to connect the Employee Import source table with the Employee Test target table. The source fields were mapped with their corresponding target fields, and the transformation was executed successfully.

The project also demonstrates the use of Coalesce to identify existing employee records based on Employee ID. New records are inserted, existing records can be updated when their details change, and unchanged duplicate data can be ignored during repeated imports.

The testing process confirmed that the configured Import Set, Transform Map, field mappings, and Coalesce settings performed according to the project requirements.

Thus, the project provides a practical understanding of **spreadsheet data import, field mapping, transformation, record validation, duplicate handling, and data management in ServiceNow**. It also demonstrates how ServiceNow can be used to efficiently transfer structured external data into a target table while maintaining data accuracy and consistency.